

Multiple Choice (10 marks)

1. If $(G, \cdot)$ is a group such that $(ab)^{-1} = a^{-1}b^{-1}$, for all $a, b \in G$, then G is a/an
 2. commutative semi group
 3. abelian group
 4. non-abelian group
 5. None of these

Find the order of the factor group $\mathbb{Z}_6 / \langle 3 \rangle$.

- (a) 2
- (b) 3
- (c) 6
- (d) 12

If $A = \{1, 2, 3\}$ then relation $S = \{(1, 1), (2, 2)\}$ is

- (a) symmetric only
- (b) anti-symmetric only
- (c) both symmetric and anti-symmetric
- (d) an equivalence relation

Let A be the set of all ordered pairs of integers (m, n) such that $7m + 12n = 22$. What is the greatest negative number in the set $B = \{m + n : (m, n) \in A\}$?

- (a) -5
- (b) -4
- (c) -3
- (d) -2

In the xy -plane, the curve with parametric equations $x = \cos t$ and $y = \sin t$, $0 \leq t \leq \pi$, has length

- (a) 3
- (b) π
- (c) 3π
- (d) $3/2$

True / False (10 marks)

Answer True or False

6. True or False: The product $(12)(16)$ in the ring $\mathbb{Z}_{24}$ is congruent to 0.
7. True or False: Every quotient ring of a commutative ring is also a commutative ring.
8. True or False: The order of a left coset aH is equal to the order of the subgroup H , which does not necessarily divide $|a|$.
9. True or False: The degree of the field extension $\mathbb{Q}(\sqrt{2}, \sqrt{3}, \sqrt{18})$ over $\mathbb{Q}$ is 4.
10. True or False: The map $x \rightarrow axa^2$ of a group G into itself is a homomorphism if and only if $a^3 = e$.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. There is an unlimited supply of congruent equilateral triangles made of colored paper. Each triangle is a solid color with the same color on both sides of the paper. A large equilateral triangle is constructed from four of these paper triangles as shown. Two large triangles are considered distinguishable if it is not possible to place one on the other, using translations, rotations, and/or reflections, so that their corresponding small triangles are of the same color. Given that there are six different colors of triangles from which to choose, how many distinguishable large equilateral triangles can be constructed? `[asy] draw((0,0)–(10,0)–(5,8.7)–cycle); draw((5,0)–(2.5,4.3)–(7.5,4.3)–cycle); [/asy]`
12. Find all values of x such that

$$\begin{vmatrix} 1 & 1 & 1 \\ x & 7 & -2 \\ x^3 & 343 & -8 \end{vmatrix} = 0.$$

Enter your values separated by commas. For example, if you think the possible values are 4, 5, and 6, then enter your answer as "4,5,6".

End of Paper